

DEEP LEARNING INDABAX SOUTH AFRICA 2022

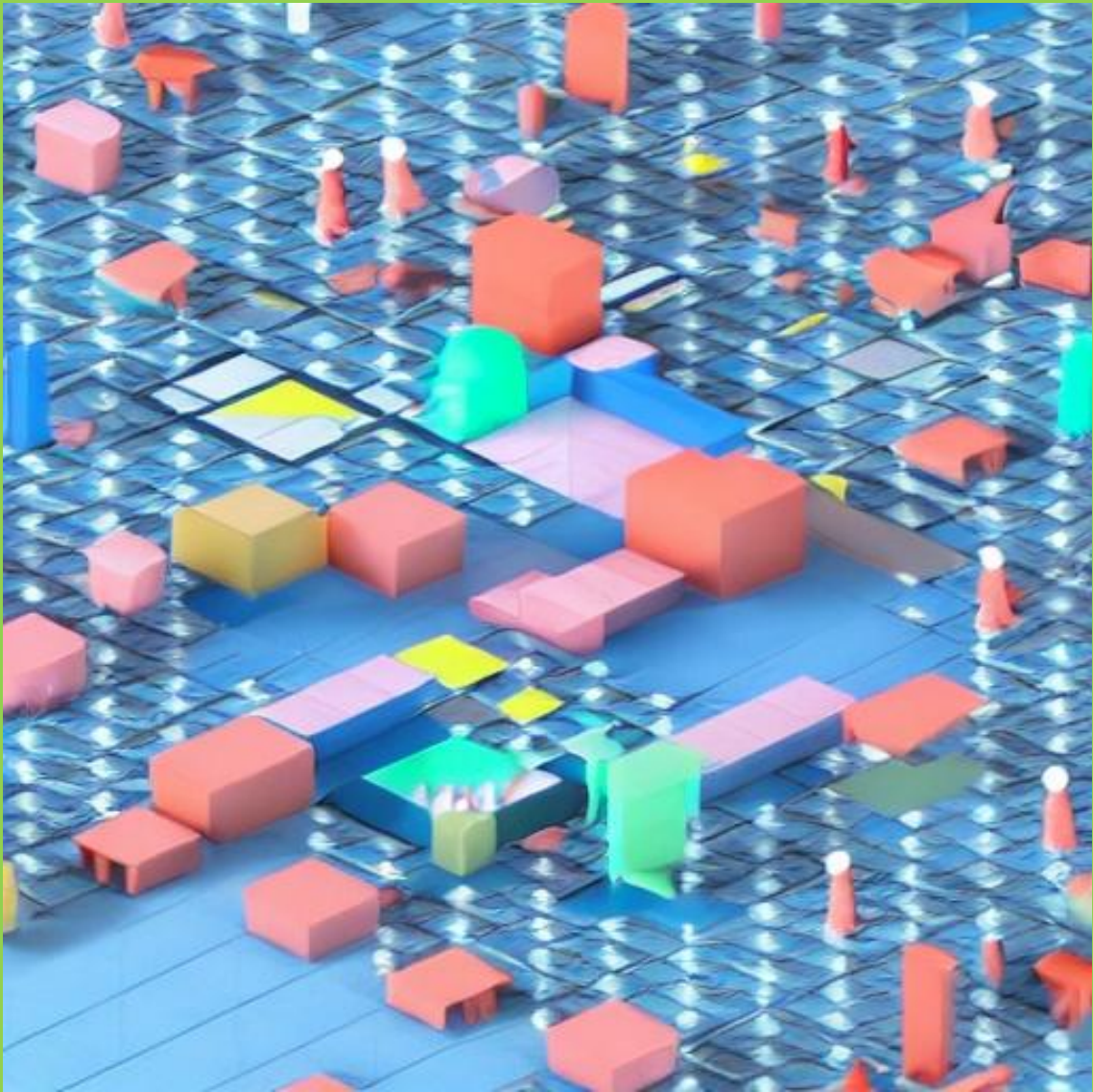
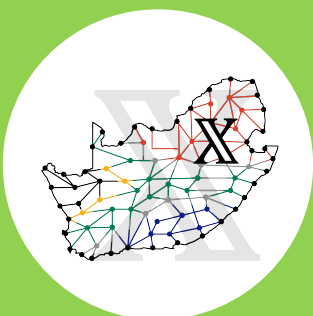


Image via: [DeepAI](#) Art Generator - word prompt: "Artificial Intelligence Conference"



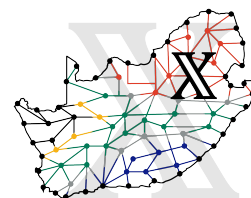
12 - 15 DECEMBER 2022

Engineering 3 Building

University of Pretoria

Pretoria, South Africa

DEEP LEARNING INDABA $\mathbb{X}$ SOUTH AFRICA 2022



ABOUT US

The Deep Learning Indaba $\mathbb{X}$ is a locally-organised meeting that helps ensure that knowledge and capacity in machine learning are spread more widely across the continent. The Indaba $\mathbb{X}$ is a means with which to experiment with the ways in which we strengthen African machine learning and its local communities, allowing more people to contribute to the conversation around machine learning and artificial intelligence. Indaba $\mathbb{X}$ events are local versions of the Deep Learning Indaba, which was held this year in Tunis, Tunisia.

LIST OF ORGANISERS

Anna Bosman

Christopher Currin

Maria Schuld

Kerryn Gammie

Lydia de Lange

Temitope Kekere

Chris Fourie

Windy Mokuwe

Dinorego Bauba Mphogo

Sicelukwanda Zwane

Guillaume Odendaal

Aboubacar Diallo

Stacey Baror

Laing Lourens

Thapelo Sindane

Avashlin Moodley

VENUE INFO

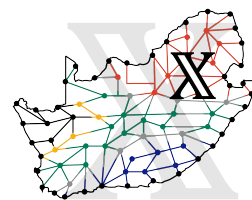
All talks take place at the **Engineering 3 Building, University of Pretoria:**

Lynnwood Rd, Hatfield, Pretoria, 0002.

The closing event will take place at the **Summit Grill & Skybar:**

Shop D(G)001, Building 5, Summit Place 221, Garsfontein, Menlyn, 0063

DEEP LEARNING INDABA X SOUTH AFRICA 2022



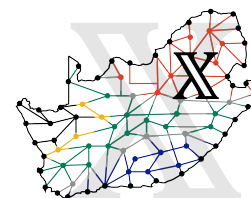
TRACKS

KEYNOTE		POSTER SESSION	
FOUNDATIONS		HACKATHON	
RESEARCH		UNCONFERENCE	
APPLIED ML & INDUSTRY		AI IN HEALTHCARE	

OVERVIEW

	12.Dec	13.Dec	14.Dec	15.Dec	
start time	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	
08h30	Arrivals	Coffee	Coffee	Coffee	
09h00		Opening Remarks			
09h30					
10h00			Break	Break	
10h30					
11h00					
11h30					
12h00					
12h30	Registration	Lunch	Lunch	Lunch	
13h00					
13h30					
14h00					
14h30		Coffee	Coffee		
15h00					Prize Giving Closing & AGM
15h30					
16h00					
16h30		Coffee	Coffee		
17h00					
17h30					
18h00					
18h30		Poster Session with AI Expo Africa	Hackathon with Zindi		Closing Event <i>buses to event available</i>
19h00					
19h30					
20h00					

DEEP LEARNING INDABA SOUTH AFRICA 2022



DAY 0

Monday, 12 Dec

ATTENDEE REGISTRATION

Engineering 3 Building, University of Pretoria

13h00 - 18h30

DAY 1 (Morning)

Tuesday, 13 Dec

MORNING COFFEE 08h30 - 09h00

OPENING REMARKS 09h00 - 09h30

KEYNOTE

Vukosi Marivate: AfricanML - Where to Next? 09h30 - 10h30

FOUNDATIONS

Anna Bosman: Fundamentals of ML and Neural Networks 10h30 - 12h00

RESEARCH

Prof Daniel Nico Wilke: Gradient-only optimization perspectives on adaptive learning rates 10h30 - 11h00

Willem van Heerdan: On the Use of Self-Organising Feature Maps as an Unsupervised Basis for Data Science 11h00 - 11h30

Jason Chalom: The Application of Attribution Methods to Explain an End-To-End Model For Self-Driving Cars (Unpublished) 11h30 - 12h00

APPLIED ML & INDUSTRY

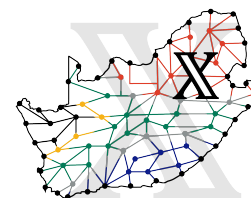
Reevana Balmahoon: Technology convergence for business transformation 10h30 - 11h00

Joanne Moonsamy: Bridging the gap between tech education and industry - addressing the missing piece 11h00 - 11h30

Naomi Molefe: AI Business - drive towards inclusion and sustainable AI businesses 11h30 - 12h00

LUNCH 12h00 - 13h00

DEEP LEARNING INDABA SOUTH AFRICA 2022



DAY 1
(Afternoon)

Tuesday, 13 Dec

KEYNOTE

Jade Abbott - Masakhane: The Unreasonable Effectiveness of Grassroots Research

13h00 - 14h00

TEA/COFFEE BREAK

14h00 – 14h30

FOUNDATIONS

Ashwinee Pandey: An Introduction to Reinforcement Learning

14h30 - 16h00

RESEARCH

Naweed Khan: Neuro-Symbolic AI: Looking beyond Deep Learning

14h30 - 15h00

David Maurice Bowman: Applications of Deep Learning to Group Theory

15h00 - 15h30

Raasetje Sefala: How else can we think about creating ground truth datasets to answer research questions?

15h30 - 16h00

APPLIED ML & INDUSTRY

Pelonomi Moiloa: Protecting Machines From Us

14h30 - 15h00

Stuart Reid: Deep volatility: uncovering material risks across 2,758,426 news stories using attention networks

15h00 - 15h30

Mahmood-Ali Parker: An Introduction to Neural Facial Authentication in the Financial Services Sector

15h30 - 16h00

TEA/COFFEE BREAK

16h00 – 16h30

FOUNDATIONS

Gemma Dawson: A Gentle Introduction to Quantum Computing

16h30 - 18h00

RESEARCH

Wiebke Hutiri: Towards trustworthy edge intelligence: lessons from voice activated services

16h30 - 17h00

Febe de Wet: Localising the Mozilla Common Voice platform for South Africa's official languages

17h00 - 17h30

Francois Meyer: Subword Segmental Text Generation for Nguni Languages

17h30 - 18h00

APPLIED ML & INDUSTRY

Cobus Louw, Louwrens Labuschagne: Sumogym Traffic Control: Reinforcement Learning From Simulation to Real World

16h30 - 17h00

Siddarth Singh: RL in Practice: Solving complex real-world problems with RL

17h00 - 17h30

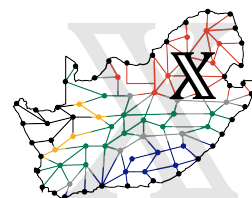
Tiffany Woodley: Reinforcement Learning: Tackling the "Curse of Dimensionality"

17h30 - 18h00

POSTER SESSION with AI EXPO AFRICA

18h00 – 20h00

DEEP LEARNING INDABA SOUTH AFRICA 2022



DAY 2 (Morning)

Wednesday, 14 Dec

MORNING COFFEE

08h30 - 09h00

KEYNOTE

Frans Cronje: How is AI changing the face of manufacturing?

09h00 - 10h00

TEA/COFFEE BREAK

10h00 - 10h30

FOUNDATIONS

Stephan Schmidt: Introduction to Bayesian Inference

10h30 - 12h00

RESEARCH

Tamlin Love: Introduction to Assisted Reinforcement Learning

10h30 - 11h00

Geraud Nangue Tasse: Towards Lifelong Reinforcement Learning through Logical Skill Composition

11h00 - 11h30

Michael Beukman: Procedural Content Generation using Neuroevolution and Novelty Search for Diverse Video Game Levels

11h30 - 12h00

APPLIED ML & INDUSTRY

Meralda Wu: From Google Collab to deployed endpoints following MLOps principles

10h30 - 11h00

Michael Williams: MLOps - You don't need it

11h00 - 11h30

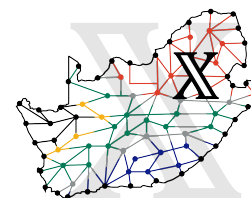
Bophelo Mkhwanazi: Test Driven AI - Lessons from Software Development

11h30 - 12h00

LUNCH

12h00 - 13h00

DEEP LEARNING INDABA SOUTH AFRICA 2022



DAY 2
(Afternoon)

Wednesday, 14 Dec

KEYNOTE

Marelle Hattingh Davel: Knowledge Discovery in Time Series Data

13h00 - 14h00

TEA/COFFEE BREAK

14h00 – 14h30

FOUNDATIONS

Ruan van der Merwe: Everything you need to know about transformers

14h30 - 16h00

RESEARCH

Dries Smit: Scaling multi-agent reinforcement learning to 11 vs 11 simulated soccer

14h30 - 15h00

Anton Lukas: Machine learning applied to pumping test analysis

15h00 - 15h30

Rirhandzu Jeaneth Novela: Temporal prediction of fine particulate matter in Mpumalanga highveld using LSTM neural networks

15h30 - 16h00

APPLIED ML & INDUSTRY

Elijah Roussos: Running Large Language models in Production for less than \$10 a month

14h30 - 15h00

Wihan Booyse: Running PyTorch in production, with Tensorflow extended

15h00 - 15h30

Nhlakanipho Kwazi Mthembu: Data Engineering with PySpark

15h30 - 16h00

TEA/COFFEE BREAK

16h00 – 16h30

FOUNDATIONS

Emile Engelbrecht: Generative adversarial networks: design, applications and dangers

16h30 - 18h00

RESEARCH

Marcel Atemkeng: Perspectives on Long-Term Research Data Archiving Solutions: the case of the SKA with learning algorithms

16h30 - 17h00

Ronald Clark: Reality Capture for Robotics, Graphics and Beyond

17h00 - 17h30

Ruan Spijkerman: Proposed Face Recognition System Based on Immune Inspired Anomaly Detection Using Symbiotic Agents

17h30 - 18h00

APPLIED ML & INDUSTRY

[MLOps] Onalerona Msimang: Error Logging — Linking Computer Science Principles with how to deal with Depression and Anxiety

16h30 - 17h00

Dr Oluwasegun Julius Aroba: The Emergent of Machine Learning and Artificial Intelligence in any relevance disciplines

17h00 - 17h30

Idris Muniru: Automatic knowledge extraction of any Chatbot through Conversation

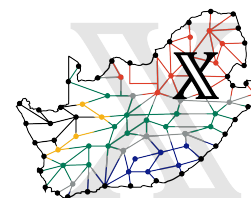
17h30 - 18h00

HACKATHON WITH ZINDI

18h00 – 20h00

DEEP LEARNING INDABA_X

SOUTH AFRICA 2022



DAY 3

Thursday, 15 Dec

MORNING COFFEE

08h30 - 09h00

PANEL DISCUSSION

Machine Learning in Research and Industry

Panelists: Nico Wilke, Jade Abbott, Marelle Davel, Joanne Moonsamy, Anna Bosman

09h00 - 10h00

TEA/COFFEE BREAK

10h00 - 10h30

HACKATHON

Zindi Hackathon

10h30 - 15h00

UNCONFERENCE

All are welcome

10h30 - 15h00

AI IN HEALTHCARE

SisonkeBiotik presentation and workshop intro

10h30 - 10h40

Joan Byamugisha: Semantic-based Document Label Verification in Biomedical Text Processing

10h40 - 11h05

Archie Arakkal: Federated learning for healthcare

11h05 - 11h30

Ifthakaar Shaik: From paper to practice: The business of AI and healthcare

11h30 - 12h00

LUNCH

12h00 - 13h00

AI IN HEALTHCARE

Greg Newman: Self-supervised Federated Learning using Wearables

13h00 - 13h30

All welcome: Ideation launch & Idea Pitch

13h30 - 14h15

Panel Session

14h15 - 15h00

CLOSING

Prize Giving for Posters & Hackathon

15h00 - 15h30

Closing Remarks & Indaba_X Annual General Meeting

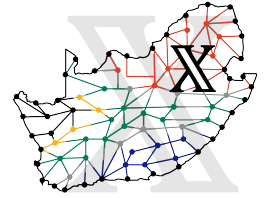
15h30 - 16h00

CLOSING EVENT

17h00 - 22h00

*Summit Grill & Skybar
buses to event available*

DEEP LEARNING INDABA SOUTH AFRICA 2022



PARTNERS



DEEP LEARNING
INDABA



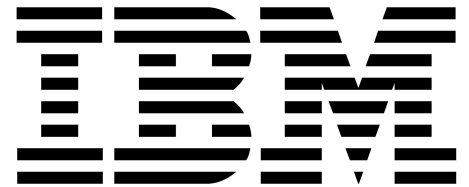
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YUNIBESITHI YA PRETORIA



M a S S
DST-NRF CENTRE OF
EXCELLENCE IN
MATHEMATICAL &
STATISTICAL SCIENCES

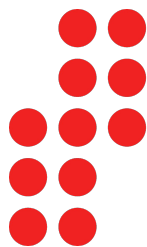
NITheCS

National Institute for
Theoretical and Computational Sciences



Google

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ASHANTI | DATA SCIENCE SPECIALIST



LELAPA AI

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